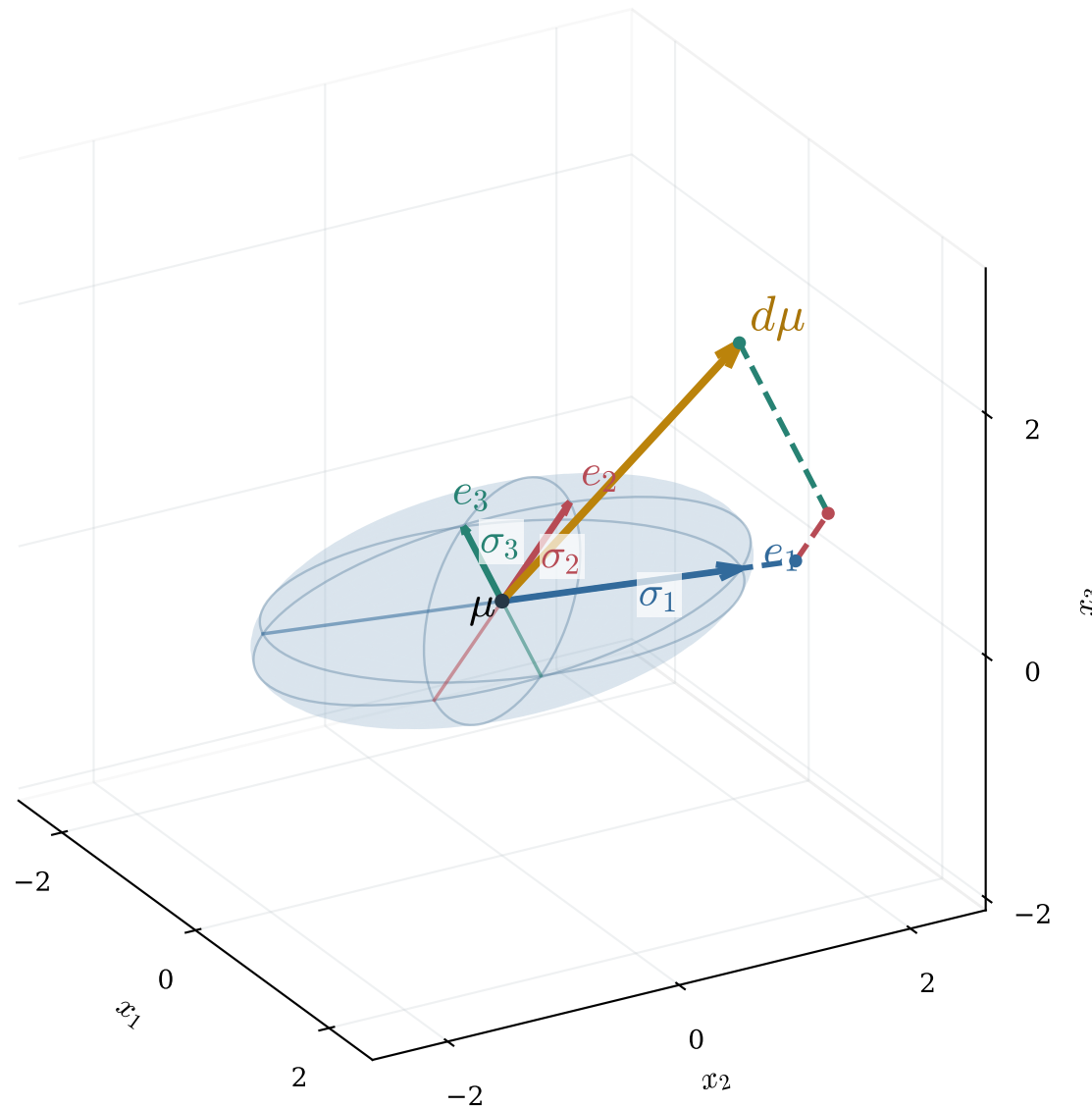


A Gaussian covariance: principal directions and scales



Covariance in its principal basis

$$R = (e_1, e_2, e_3), \quad R^T R = I$$

$$D = \text{diag}(\sigma_1^2, \sigma_2^2, \sigma_3^2)$$

$$\Sigma = R D R^T$$

The same mean tangent, in this basis

$$R^T d\mu = (e_1^T d\mu, e_2^T d\mu, e_3^T d\mu)^T$$

$$d\mu = \sum_{i=1}^3 (e_i^T d\mu) e_i$$

Dashed segment 1: $(e_1^T d\mu)e_1$

Dashed segment 2: $(e_2^T d\mu)e_2$

Dashed segment 3: $(e_3^T d\mu)e_3$

Components in the current eigenvector basis.
The vector itself is unchanged.

$$\text{Constant-density surface: } (x - \mu)^T \Sigma^{-1} (x - \mu) = 1$$